Comments on Mechanism Questions #0

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle* and edited in *Affinity Designer*.

Here I present some comments on the problems of the set handed out in our first class. There are too many questions in any one set to complete in 50 minutes. Often they will be left for you to complete on your own. For the first few classes, I will provide these comment documents because the foundation of organic chemistry is very important.

1. When using isotopically labeled water, the hydrolysis of epoxide gives different products depending on if the reaction is performed in acidic or basic solution. Can you explain the observation by proposing a mechanism for each reaction? How would you exploit this observation in organic synthesis?

The epoxide is asymmetrical. One carbon is secondary and the other is tertiary. In base catalyzed ring opening the hydroxide nucleophile cannot attack at the tertiary centre. Hydroxide attacks at the secondary carbon in a S_N2 substitution reaction. The alkoxide product can then accept a proton from water to regenerate the hydroxide. If the hydroxide that attacked contained an isotope of oxygen, we would see that isotopic label occur at the secondary carbon in the diol product only.

In acid, there are no strong nucleophiles present. The small amount of protonated epoxide can open to relieve ring strain. The tertiary carbocation is much more likely to form than the secondary carbocation, so we will see the vast majority of intermediate be the tertiary cation. Water solvent, which is a weak nucleophile, can add to the strongly electrophilic carbocation. As a result the majority of product would see an isotopically labelled oxygen at the tertiary centre of the diol product.

In a synthesis, we could control where a nucleophilic oxygen added by performing the reaction in acid or in base. Controlling regioselectivity in a reaction is a major goal of synthetic organic chemistry and we will master these principles in this course.

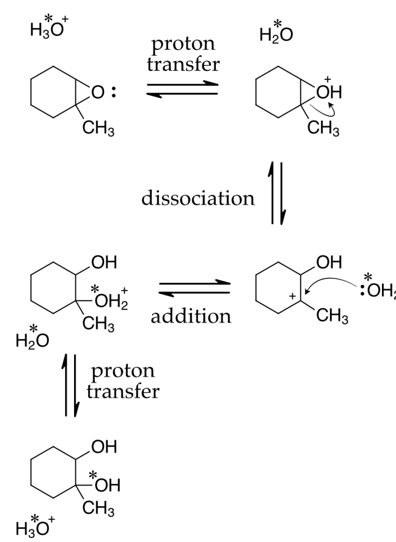

Figure 1: Acid catalyzed epoxide opening.↑

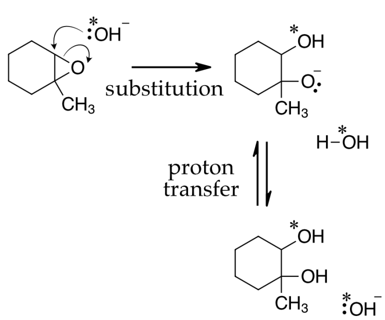

Figure 2: Base catalyzed epoxide opening. ⟵

2. Explain the mechanisms that give rise to each of the products below. What reaction conditions would you choose to maximize the yield of each?

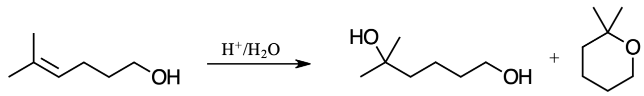

Figure 3: Acid catalyzed addition to an alkene.
⟵

The acid could protonate the alkene or the alcohol. Protonating the alcohol will lead nowhere because we cannot dissociate the water to create a primary carbocation. Always remember your vow to never propose an unstabilized primary carbocation. Protonating the alkene would give a tertiary carbocation that can have water add to give the alcohol product or have intramolecular ring closing to give the cyclic ether.

To maximize the yield of the diol we should perform the reaction in water. To maximize the yield of the cyclic ether we should perform the reaction in organic solvent using a non-aqueous acid such a sulphuric acid or TsOH

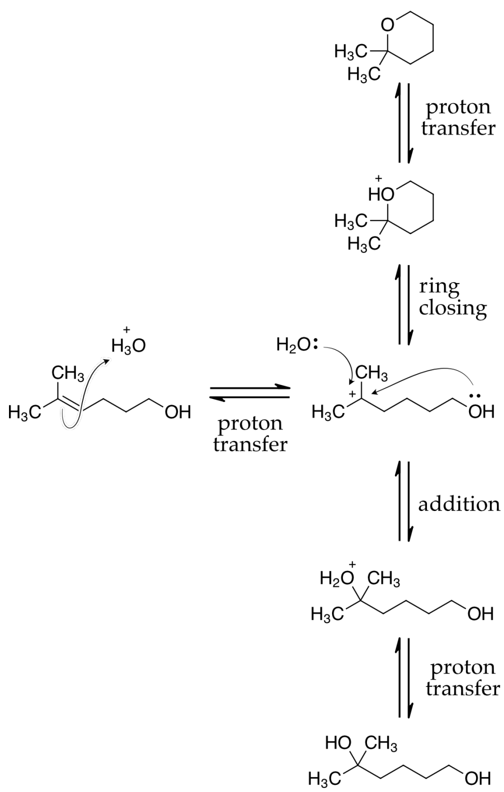

Figure 4: Acid catalyzed addition to an alkene.
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3. Bromination of acetone in water is fast but the rate is independent of bromine concentration. (See figure 5). Can you explain the observation by proposing a mechanism for the reaction?

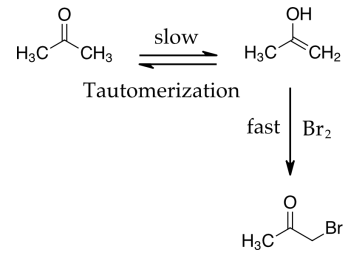

Figure 5: α-bromination of acetone.↑

Bromine reacts rapidly with alkenes. The reactive species in this reaction is the enol tautomer of acetone. Enols are electron-rich and will react rapidly with bromine. It is the formation of the enol that is the rate-determining step followed by the rapid bromination reaction. So the rate law is zero order in bromine concentration.

The first step is the tautomerization to form an enol. The equilibrium constant for the enol/ketone equilibrium of acetone is 8×10^{-8}. Only a small amount of the enol will be present, but it is an electron-rich alkene and reacts with the bromine quickly. The rate of formation of the enol is often the rds.

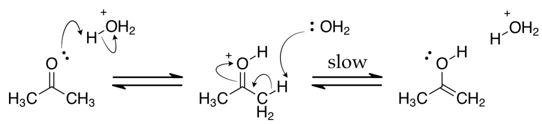

Figure 6: Tautomerization of a ketone to form an enol. ⟵

The concentration of bromine does not affect the rate in water. Take note of the fact that the rate of α-halogenation is the same whether chlorine, bromine or iodine is used as the reactant.

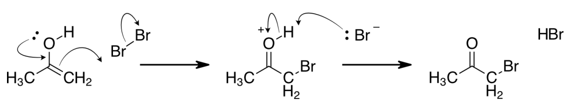

Figure 7: Electrophilic bromination of the electron-rich double bond of the enol. ⟵

4. The benzoin condensation seems to be impossible. And it is unless catalyzed by cyanide ion. Propose a mechanism. Be able to argue for the stability of your intermediates.

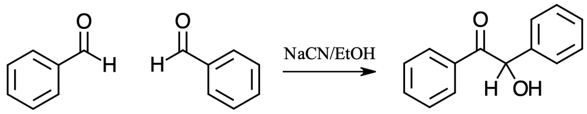

Figure 8: The benzoin condensation. ⟵

You may think that the benzoin condensation can be explained by the cyanide anion acting as a base to deprotonate the carbonyl group of the benzaldehyde and make a nucleophile as shown in figure 9. The anion that we imagine this way is a very high energy anion. The lone pair is not stabilized and this species is not possible (at least it has never been observed.) Yet we observe reactions like this in biochemistry all the time.

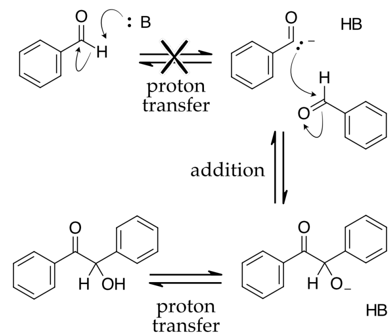

Figure 9: An impossible mechanism.↑

The mechanism is shown in figure 10. The cyanide ion adds to the carbonyl group of the aldehyde. The adduct can be deprotonated because the resulting negative charge can be delocalized through the π system of the cyano group (figure 11.) We do not have a deprotonated aldehyde, we have a deprotonated cyanohydrin. This is the nucelophile that can add to another benzaldehyde to give an aldol product. The cyano group is finally eliminated to regenerate the carbonyl group and the cyanide anion. The cyanide is a catalyst.

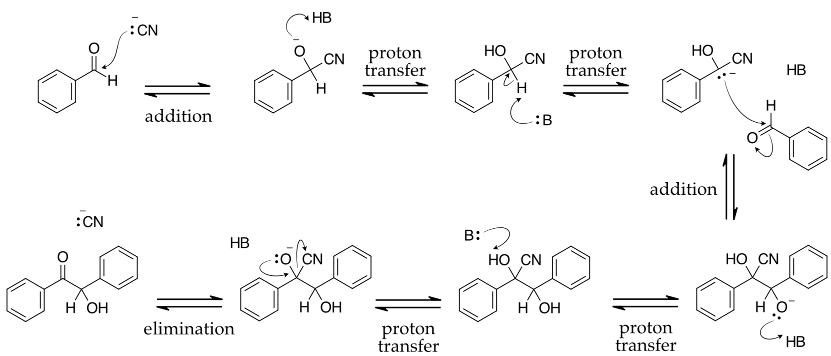

Figure 10: The mechanism of the cyanide-catalyzed benzoin condensation. ⟵

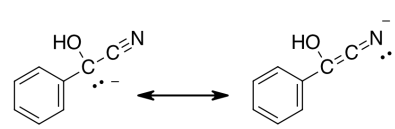

Figure 11: Stabilization of the cyanohydrin anion.↑